

Day 9 — Mixed Practice + Project Session

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will complete a mixed-practice drill covering 2-digit $\times 11$, 3-digit $\times 11$, and $\times 12$ – $\times 19$ problems, and will make substantial progress on their project (video, workbook, or proof poster).

Materials

- Mixed-practice drill sheet (20 problems).
- Project materials per pair: paper, markers, phone (for video pairs), template sheets (for workbook pairs).
- `assessments/rubric.md` posted on the wall for reference.

Lesson flow

Warm-up (3 min)

- 8 quick mental $\times 11$ problems including carry. Should be near-instant.

Mixed-practice drill (12 min)

- Distribute 20-problem sheet:
 - 5 problems: 2-digit $\times 11$ (no-carry)
 - 5 problems: 2-digit $\times 11$ (carry)
 - 5 problems: 3-digit $\times 11$
 - 5 problems: 2-digit $\times 12$ through $\times 19$
- Solo, untimed, but aim to finish in 10 min.
- 2 min: pair-swap, peer-check three problems each.

Project session (28 min)

- Project format options (pick on Day 8 or earlier):
 - **Format 1: 60-second teaching video** for a Grade 4 audience.
 - **Format 2: Printed workbook** — 25 problems organised in difficulty levels, with worked examples and an answer key.
 - **Format 3: Proof poster** — one A3 sheet showing the algebraic derivation of $(10a + b) \times 11 = 100a + 10(a+b) + b$ with examples and a 3-line explanation.
- Each pair shows the teacher their plan in the first 5 min. Teacher signs off or redirects.
- Work time. Teacher circulates:
 - Check video pairs have a quiet recording space.

- Check workbook pairs have grid paper.
 - Check poster pairs have A3 sheets and at least 2 markers.
- Final 5 min: each pair shares their working title and ONE thing they're stuck on. Class offers one suggestion.

Wrap-up (2 min)

- *“Tomorrow is the final assessment and project share. Bring your finished artefact. Polish, don't restart.”*

Student-facing content

Today's two jobs

1. **Drill.** 20 mixed problems. Self-assess.
2. **Project work.** Make real progress today — by end of class you should be 70% done.

Project Day 10: 2-minute presentation + 1-min Q&A. Be ready.

Homework

- Finish your project at home. 30 minutes max. If it's not finishable in 30 minutes, **simplify it** — better a complete simple project than a half-done ambitious one.

Differentiation

- **Tier up:** Add a “challenge” section to your project: a 4-digit $\times$ 11 worked example with cascading carries.
- **Tier down:** Use the workbook template provided. Don't custom-design from scratch today.

Teacher notes

- The project here is intentionally smaller-scope than Module 5's, because the technique is narrower (just $\times 11$ plus the $\times 12$ – $\times 19$ family) and the assessment is more on the algebra than on the artefact.
- Watch for pairs who haven't started by mid-class. Pull them aside; offer the simplest format (proof poster) as a fallback.
- The proof poster format is recommended for pairs that struggled on Day 3 — making the algebra public forces them to internalise it.

Citation(s) used in this lesson

- (No new citations — application day.)